Robotic Exoskeleton for Load Transportation

IMU-based locomotion intent recognition and assist gating in MATLAB

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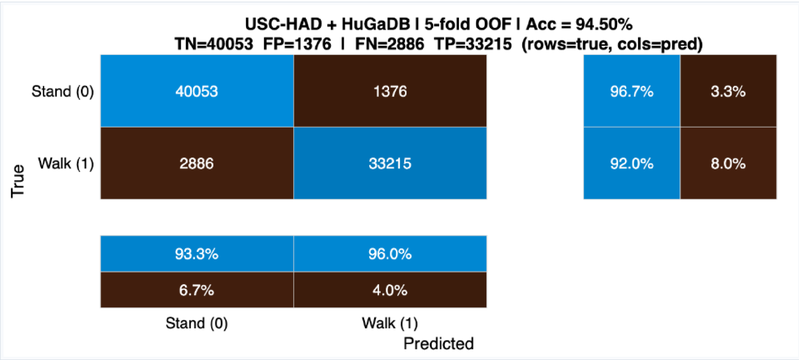
Key message: a reproducible MATLAB pipeline combines Kalman fusion, binary SVM classification, and FSM hysteresis to turn noisy IMU data into a stable assist on/off command.

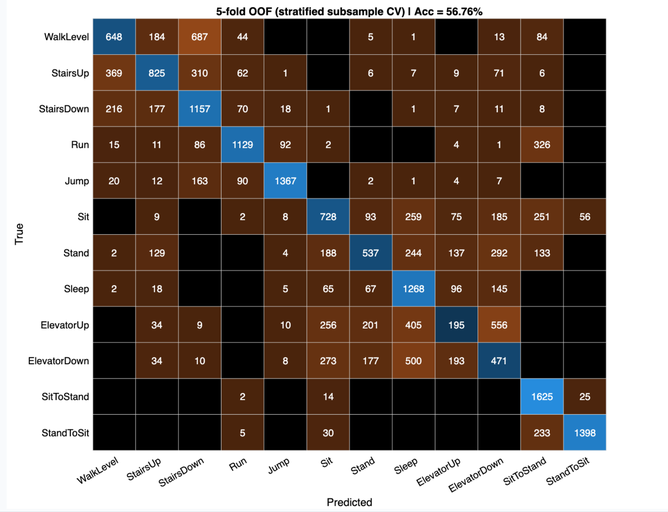
Deployment baseline
Binary SVM, merged 5-fold OOF
94.50%
Best validated claim for assist gating

Exploratory activity model
12-class ECOC, 5-fold OOF
56.76%
Useful for analysis, not direct control

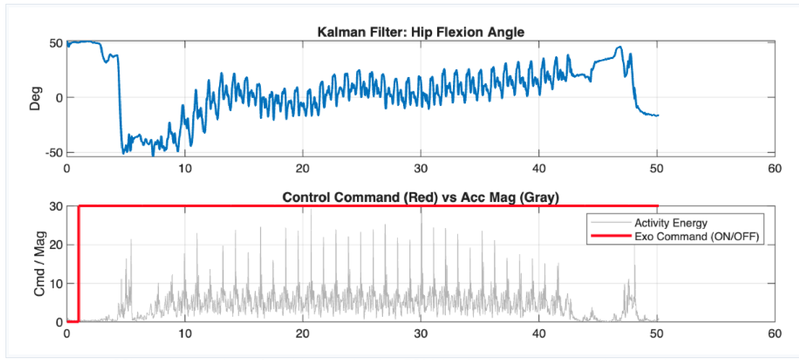
Sequence baseline
Binary LSTM, 20% holdout
96.71%
Separate protocol, not directly comparable



Merged binary SVM: strong balance between missed locomotion and false activation



12-class ECOC: much harder task, with confusion across similar activities



CSV replay pipeline: fused hip angle plus FSM-latched assist command in simulation

Why this project matters

Workers carrying loads above 20 kg face high fatigue and musculoskeletal risk. For an exoskeleton to help, assistance must switch on when locomotion begins and switch off when the user stops, without invasive sensors or constant manual tuning.

This project therefore focuses on a **software supervisor** rather than a full torque controller: detect locomotion-like intent from IMU data, estimate motion state, and issue a reliable assist gating signal.

Delivered pipeline

- **Inputs:** public IMU datasets plus a three-phone CSV replay prototype.
- **Fusion:** Kalman-based hip-angle estimation using `imufilter`.
- **Windows:** 1.0 s at 100 Hz with 50% overlap.
- **Features: 30-D** vector, five statistics per IMU slot across six slots.
- **Classifier:** RBF SVM for binary locomotion-like vs. non-locomotion.
- **Supervisor logic:** FSM hysteresis, 3 consecutive positives to engage assist and 5 negatives to release.

Data and feature design

HuGaDB contributes six on-body IMU streams per window. USC-HAD contributes one IMU stream, with the remaining slots zero-padded so both datasets share the same feature layout.

That design makes the **merged** model more realistic than the USC-only score: it sees broader sensing layouts and more heterogeneous activity labels.

Validated quantitative result

The strongest deployable claim is the **merged binary SVM**: it reaches **94.50%** five-fold out-of-fold accuracy while preserving a reasonable trade-off between false assist activation and missed locomotion onset.

Training data	OOF acc.
USC-HAD only	**98.91%**
HuGaDB only	**88.60%**
USC-HAD + HuGaDB	**94.50%**

The USC-only score is numerically highest, but the merged model is the better poster result because it generalizes across a broader sensing setup.

How to read the other models

12-class ECOC (56.76%): useful as an exploratory activity-recognition baseline, but not yet strong enough for fine-grained control decisions.

Binary LSTM: holdout accuracy is **96.71%**, but it was evaluated on a stratified 20% split rather than five-fold OOF. It therefore remains a **separate baseline**, not evidence that it outperforms the SVM.

Controller behavior

The pipeline figure shows why the FSM matters: even with a high-accuracy classifier, raw predictions can chatter at transition boundaries. Hysteresis converts those predictions into a smoother and more usable assist signal for simulation-driven testing.

Take-home message

A reproducible baseline for binary assist gating has been delivered.

The repository now supports public-dataset evaluation, CSV replay, and stable assist-command generation in MATLAB.

What is not claimed yet

- No hardware-in-the-loop or human-subject trials under load.
- The 100–200 ms predictive horizon remains a **design target**, not a re-benchmarked final result.
- Current output is **assist gating**, not full torque scheduling or terrain-aware control.
- LSTM and SVM have not yet been compared under the same validation protocol.

Next steps

- Align SVM and LSTM evaluation under the same cross-validation protocol.
- Add hardware integration, latency measurement, and load-carrying trials.
- Move from binary assist gating toward subject-specific, activity-aware assistance.

Code and report:
https://github.com/Rex-Yim/AutomationForExoskeleton